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Principles of Information Visualization

Assignment 1

- **Reading:** Graphics Lies, Misleading Visuals
 10 min
- **Peer-graded Assignment:** Graphics Lies, Misleading Visuals
 1h
- **Review Your Peers:** Graphics Lies, Misleading Visuals

Peer-graded Assignment: Graphics Lies, Misleading Visuals

It looks like this is your first peer-graded assignment. Learn more

You passed!
Congratulations! You earned 15 / 15 points. Review the feedback below and continue the course when you are ready.

Instructions **My submission** **Discussions**

Some of your fellow learners have submitted their reviews anonymously. All names are still visible to course instructors.

Cherry-Picked Climate to prove no global warming

Submitted on September 20, 2024 Shareable Link

PROMPT

Read Alberto Cairo's work, Graphics Lies, Misleading Visuals

Locate an example of a misleading visual that uses one or more of the mechanisms for misleading that Cairo outlines in his book chapter: (1) Hiding relevant data; (2) Displaying too much data and obscuring reality; (3) Distorting data through visual forms.

Please upload an image of this visual using a widely accessible graphic format (e.g., PDF, .jpg, .png)

Cherry-Picked Climate to prove no global warming



Cherry-Picked Climate to prove no global warming

RUBRIC

Has the learner uploaded a visual?

- 0 points — No
- **1 point** — Yes

BL DP

PROMPT

Briefly describe the context for the visual by addressing the following questions:

1. What is the source of the visual? (e.g., URL or bibliographic citation)
2. Who is the intended audience (i.e., decoders)? How do you know this?

1. **Source of the visual**
https://www.visilies.org/2017/gallery/
https://drive.google.com/file/d/13GWDrfjUbSRQutLkrj1GV_guuQML5puV/view

2. **Intended audience**
The intended audience for Monckton's visuals is primarily climate change skeptics, policy-makers, and the general public who are not well-versed in climate science. This can be inferred based on several factors:

- Political and Ideological Context: Monckton often speaks at conferences, writes articles, and appears in media outlets that cater to conservative, right-leaning, or climate change-denying groups. His presentations are designed to appeal to people who are skeptical of mainstream climate science, either due to political ideology or economic interests (such as those in the fossil fuel industry).

- Simplified and Manipulative Visuals: The visuals Monckton uses tend to distort data in ways that are difficult for non-experts to detect. The use of truncated axes, selective time frames, and misleading representations are all targeted at an audience that may not have the scientific literacy to spot such manipulations. This suggests that the target audience is not climate scientists, who would be quick to critique the inaccuracies, but rather the general public or decision-makers with limited knowledge of scientific data visualization techniques.

- Public Speaking and Media Appearances: Monckton's outreach efforts are aimed at influencing public opinion and political decision-making, as seen in his appearances before government bodies and in debates about climate policy. His rhetoric is often framed in a way that appeals to emotions and pre-existing beliefs about government overreach, economic consequences, or skepticism of scientific authority, further indicating that his audience is broad and includes individuals who may already be inclined to question the reality of climate change.

RUBRIC

Did the learner provide the source of the visual?

- 0 points — No
- **1 point** — Yes

BL DP

Did the learner identify the intended audience? Did they provide an explanation of how they know who the intended audience is?

- 0 points — The learner did not identify the intended audience.
- 2 points — The learner identified the intended audience, but did not provide an explanation.
- **4 points** — The learner identified the intended audience and they provided an explanation of how they knew who the intended audience is.

BL DP

PROMPT

1. Identify the specific component(s) of the visual that is/are misleading
2. For each part(s) of the visualization that is/are misleading, identify the mechanism that is used: hiding relevant data to highlight what benefits us; displaying too much data to obscure reality; using graphic forms in inappropriate ways (distorting the data)
3. Explain how the mechanisms are used to mislead

1. **Specific component(s) of the visual that is/are misleading**:

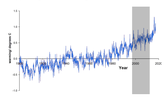
One common misleading component in Monckton's visuals is the **selective use of data**. For example, he often highlights specific time periods or datasets that support his claims while ignoring broader trends that contradict them
.

2. **Identification of the mechanism used for misleading visuals**

The graph plotted in the image attached shows the average global temperature over a course of 15 years, from 1997 to 2012. The goal of this graph is to prove that global warming is not happening. i.e., seems to invalidate claims that global temperatures are rising.

The plot looks legit, and the data behind it are in fact **accurate**. But the problem with the graph is it's **cherry picking data**. That is, this graph is showing a subset of available data that supports a desired conclusion and is throwing away data that does not support that conclusion. In the grand scheme of things, 15 years is not very long. What happens if we expand the time frame.

If one expands the data from 1900 to 2017. The shorter 15 year period from the first graph is highlighted in the gray box. As you can see, if we choose to look only at that specific 15 year period, the average global temperature did hold steady. But the decades before this region saw a noticeable average increase in temperature and the years following this region also show rising temperature. When shown in a larger context, the time frame from 1997–2012 does not accurately represent the larger trend, as shown below from the next plot, which shows there is a clear increasing trend in the global temperature (supports **global warming**).



- **Selective Data Presentation**: This involves **hiding relevant data (only selecting recent 15 years)** to highlight what benefits his argument.
 By cherry-picking data, Monckton can create a misleading narrative that appears scientifically valid
 1
- **Inappropriate Graphic Forms**: Monckton sometimes uses graphs and charts in ways that distort the data.
 For instance, he might manipulate the axes or scales to exaggerate certain trends
 2

The primary reason for misinterpretation in Christopher Monckton's climate visuals is the **distortion of data through graphic forms**, particularly **manipulating the Y-axis** and **selective data presentation**. Monckton is known for using charts that intentionally cherry-pick data or truncate the Y-axis to minimize the appearance of global warming trends. By showing only specific time periods, such as a temporary cooling phase, or by choosing a scale that minimizes the visual impact of rising temperatures, Monckton creates a misleading impression that global warming is either negligible or not occurring.

Additionally, **hiding relevant data** is another mechanism Monckton uses. For instance, he may exclude data from more recent years or neglect to show long-term climate trends, focusing instead on short-term variations in temperature. This selective presentation of information skews the narrative, allowing him to argue against the overwhelming scientific consensus on climate change.

These tactics contribute to a **false narrative** that global warming is a minor or temporary phenomenon, despite the broader data showing clear and sustained increases in global temperatures.

3. **Explanation of how the mechanisms are used to mislead**

- **Selective Data Presentation**: By only showing data from specific short time periods (**only for recent 15 years**), Monckton can make it seem like there is no significant warming trend.
 This selective presentation hides the overall upward trend in global temperatures, misleading viewers into thinking that climate change is not a serious issue
 1.
- **Inappropriate Graphic Forms**:
 Manipulating the axes or scales can make minor fluctuations appear more significant than they are. For example, in the above plot, the Y axis chooses the unit in **10th of degrees above and below 14 degree Celsius** (instead of choosing the actual degrees) - which is an example of **distortion of data through graphic forms**, particularly **manipulating the Y-axis.**
 This can lead viewers to draw incorrect conclusions about the severity or causes of climate change
 2.

RUBRIC

Did the learner identify one or more specific components that is/are misleading?

- 0 points — No
- **3 points** — Yes

BL DP

For each component of the visual that the learner identified as misleading, did the learner identify the mechanism that was used to mislead?

- 0 points — No
- **3 points** — Yes

BL DP

Did the learner explain how the mechanisms were used to mislead?

- 0 points — No
- **3 points** — Yes

BL DP

PROMPT

Optional: Describe any additional issues you found with visual that did not fall under Cairo's three misleading mechanisms.

Misinterpretation of Scientific Studies:

Monckton often misinterprets or misrepresents the findings of scientific studies. He might cite a study to support his claims, but upon closer inspection, the study actually contradicts his argument1.

Use of Non-Peer-Reviewed Sources:

He frequently relies on sources that have not undergone rigorous peer review, which undermines the credibility of his claims (refer to https://skepticalscience.com/docs/Monckton_vs_Scientists.pdf).
|
These tactics collectively contribute to spreading misinformation about climate change, making it harder for the public to understand the true nature and urgency of the issue.

RUBRIC

Optional: If the learner described additional issues and you would like to provide feedback on their comments, please do so here.

Bruno Lurner
Well done

Translate to English

David Jonathan Perlick

Translate to English

good

Translate to English

Start new attempt

Comments
Comments left for the learner are visible only to that learner and the person who left the comment.

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